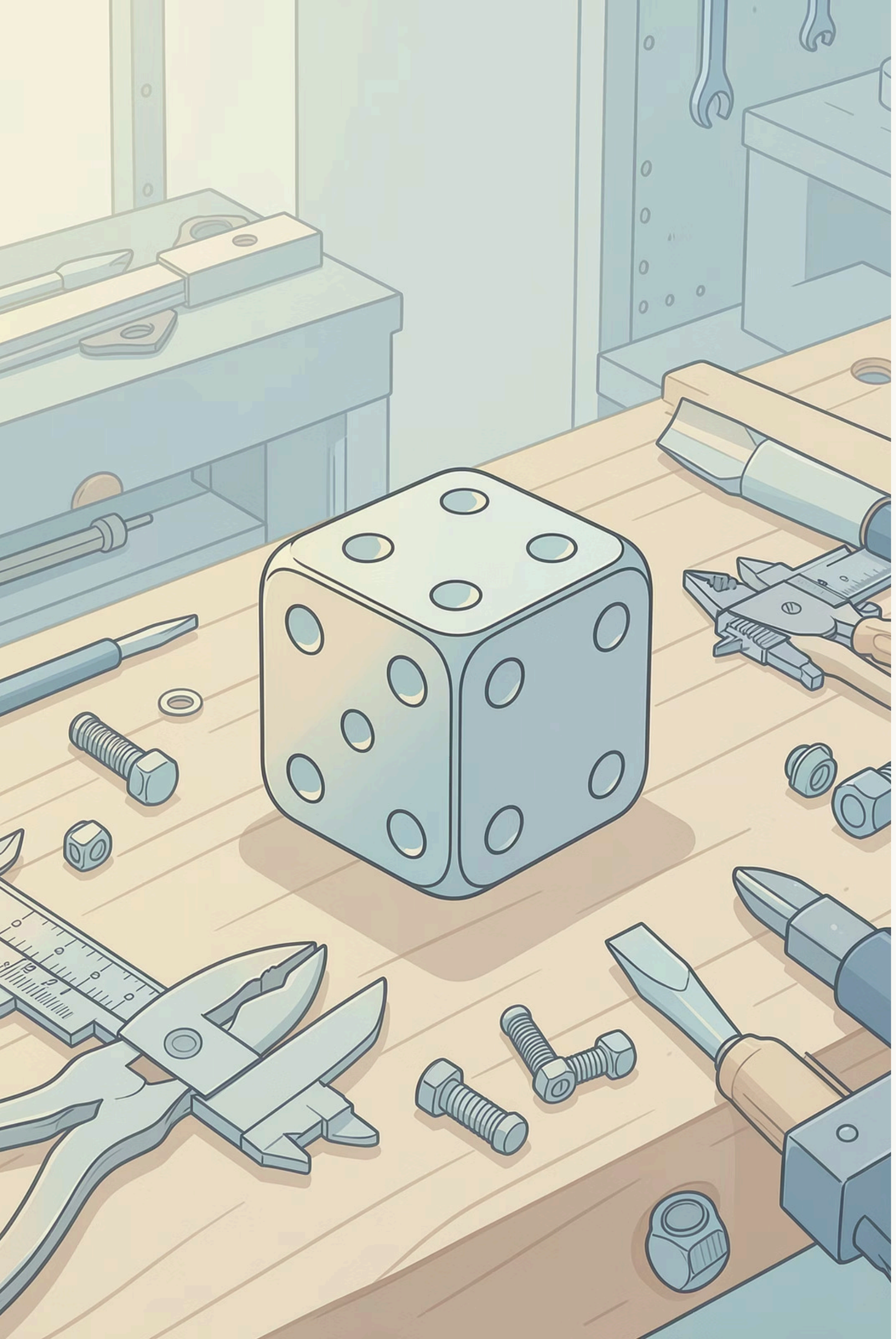

Dice by Hand Tools

A step-by-step journey in precision metalwork — from raw mild steel block to a finished dice, crafted entirely by hand.

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TMJ 3MO-2 / TMJ MR2

Materials & Tools

What We Used

Material

Mild steel block

Measuring

Ruler, calipers, scriber & square

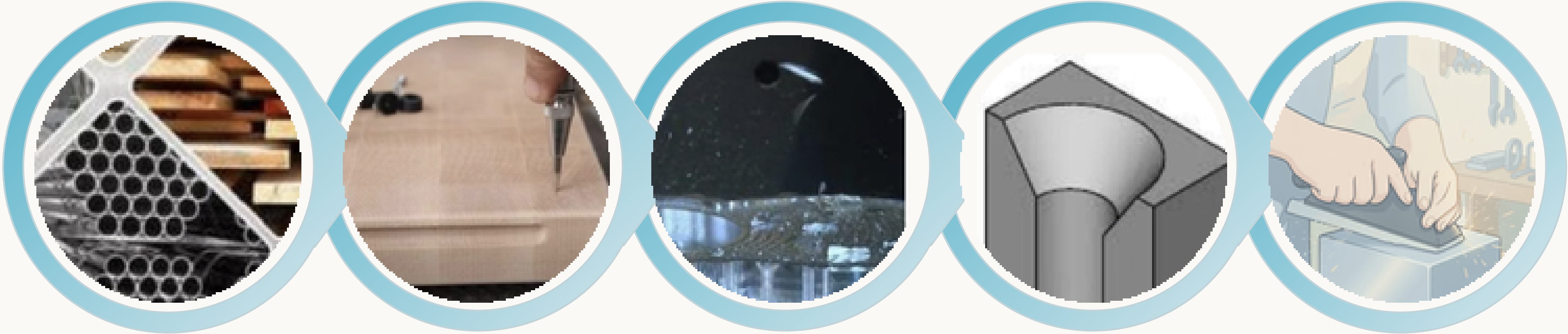
Marking

Center punch & hammer

Shaping

Drill, drill bits, countersink bit, file & sandpaper

The Process at a Glance



Each step builds on the last — precision at every stage is what separates a good dice from a great one.

Measuring & Layout

→ **Equal Spacing**

Measured all faces for uniform proportions.

→ **Grid Lines**

Drew precise lines using a square and scriber.

→ **Pip Positions**

Marked exact dot locations on each face.

Centre Punching

Why It Matters

A centre punch creates a small indent at each marked location, ensuring the drill bit seats correctly and does not wander or slip during drilling.

Accuracy at this stage directly determines the quality of the final pips.

Drilling

Controlled Drilling

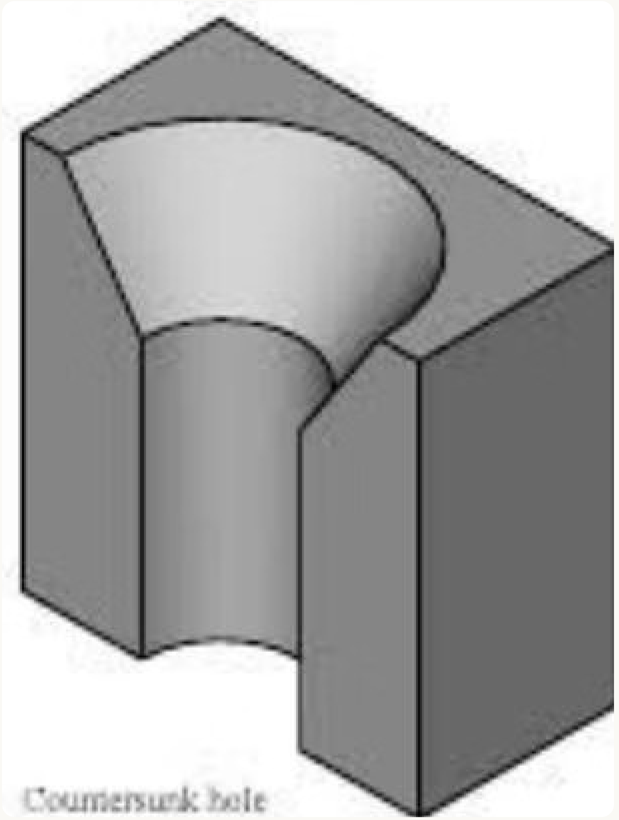
Holes were drilled at each punched point, keeping the drill perfectly straight and applying steady, controlled pressure throughout.

⚠ A wandering drill bit can ruin the layout — patience and control are essential.

Countersinking the Pips

Creating the Pips

A countersink bit was used to widen and shape each hole into a clean, rounded pip. Consistent depth across all holes ensures a professional, uniform look.

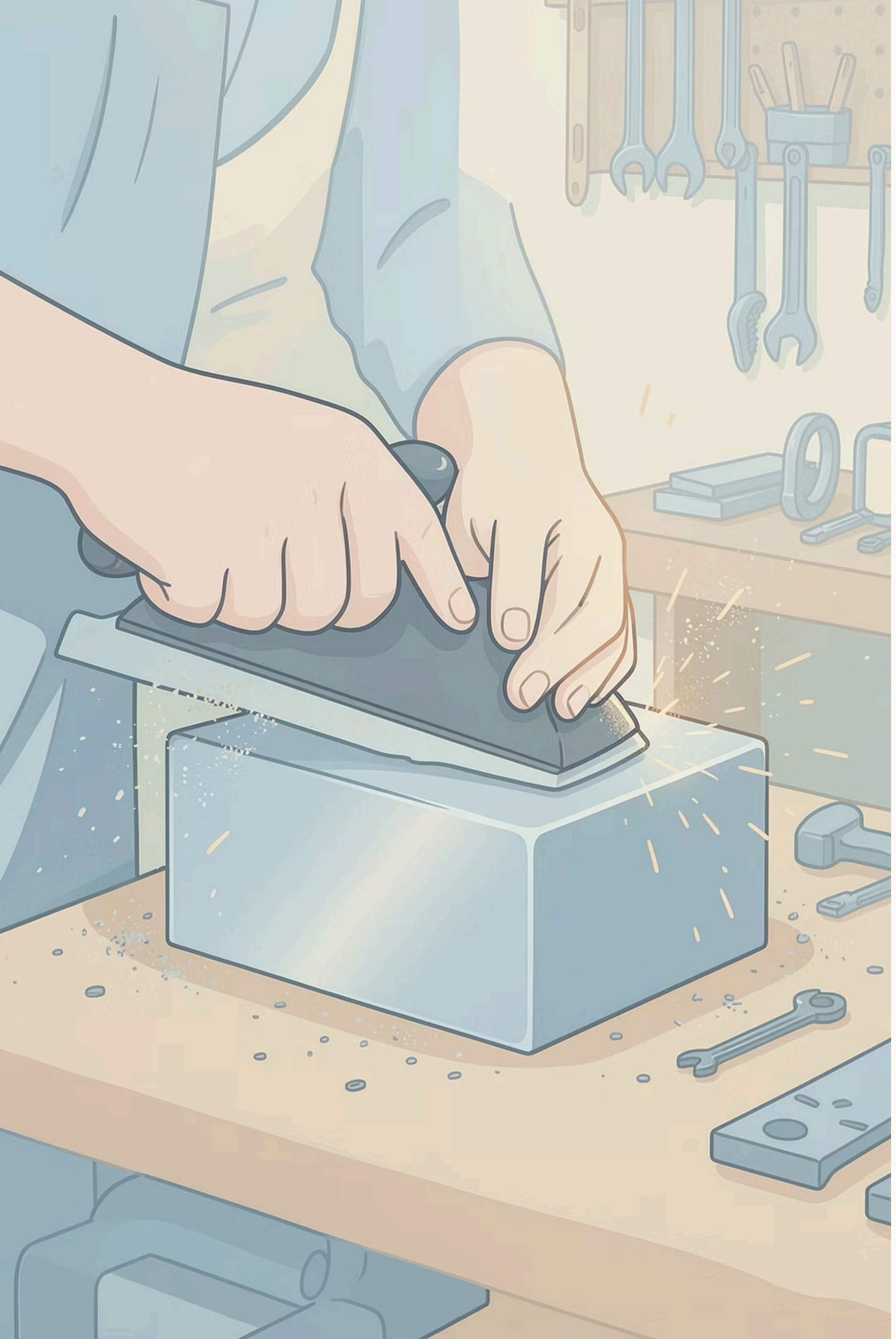


Countersunk

Conical recess — used for the dice pips

Counterbored

Flat-bottomed recess — reference comparison



Filing & Finishing

Filing

Edges filed to remove burrs and achieve a smooth, even shape.

Sanding

Surfaces sanded progressively for a cleaner, refined finish.

The Final Product

Completed entirely with hand tools — each face showing precise pip placement and clean countersunk holes.

Face: 6

Single centred pip — clean and precise.

Face: Three

Three pips in a diagonal — evenly spaced.

Face: Four

Quincunx pattern — most complex face.

Reflection

What Went Well

Accurate drilling and clean layout — the grid lines held up throughout the process.

Challenges

Keeping pip spacing perfectly even and drilling consistently straight were the toughest parts.

Improvements

Better surface finishing and deeper, more uniform countersinks would elevate the final result.